
MineOcclude: A Paired Visual-Evidence Benchmark for Object-Position Decoding

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Abstract

1 How accurately can object position be decoded from a pixel-based world model
2 as visual evidence is reduced? MineOcclude pairs the same realized Minecraft
3 trajectory across no wall, three stained-glass materials, and an opaque wall, using
4 renderer-defined visibility and temporally aligned server-state targets. We select a
5 spatial readout family on separate development data and evaluate frozen MineWorld
6 activations on 25 new identities, with grouped nested validation and probes fitted
7 separately within each condition. Mean trajectory RMSE is 0.079–0.102 blocks in
8 the visible and glass conditions and 1.645 blocks behind stone. The brown-glass-
9 to-stone increase is 1.561 blocks, with a 95% identity-bootstrap interval of [1.466,
10 1.652]. At complete opacity, the neural readout has lower error than a linear current-
11 frame pixel baseline but higher error than direction-and-age and oracle-history
12 predictors. Neither of two temporal transformations provides supported average
13 improvement across the partial and opaque conditions. The protocol separates
14 visual evidence, readout accuracy, and trajectory predictability. Its present results
15 characterize separately fitted linear probes of a reset 16-frame window under
16 retained BGR preprocessing, not continuous hidden-state tracking or intended-
17 RGB behavior.

18 1 Introduction

19 Agents must often estimate hidden state from past observations [1–3]. World models can support this
20 through learned dynamics and memory [4–6]. Interactive models generate visual continuations from
21 pixels and actions [7–11], but plausible outputs do not establish which object states remain internally
22 accessible [12, 13].

23 Occlusion separates current sensory evidence from prediction based on history or dynamics [14–18].
24 Visible position can be recovered from an image [19, 20]; hidden-position estimation instead depends
25 on prior observations or predictive cues [21–24]. Behavioral benchmarks assess prediction or physical
26 plausibility [25–27], while probes test accessible state [28–31]. Decodability alone does not establish
27 calibration [32, 33], object-centric structure [34, 35], or causal use [36–40]. We ask how accurately
28 aligned object position remains linearly accessible as visual evidence is reduced.

29 MineOcclude pairs the same sheep motion, camera, and actions across visual conditions. Server state
30 supplies wall-parallel world- x targets, aligned to rendered motion with causal finite-impulse-response
31 (FIR) filters. We first use an earlier wall/no-wall cohort to select a spatial readout. Our central
32 study, Experiment III-A, then tests that readout family on 25 new identities across no wall, three
33 stained-glass materials, and an opaque wall. The studies use separate cohorts and are not pooled.

34 Our main contribution is the paired material/evidence protocol with continuous aligned targets,
35 demonstrated on one checkpoint and apparatus. Grouped evaluation separates development-time
36 readout selection from the new-cohort comparison. Pixel, metadata, and oracle-history predictors
37 distinguish readout accuracy from task predictability; the study does not establish a new memory
38 mechanism or a superior state estimator. The anonymized project entrypoint is our [anonymous](#)
39 [repository](#).



Figure 1: Illustrative no-wall and opaque-wall captures. Matched trajectories separate object motion from visual evidence. The central study adds three stained-glass conditions; these two images illustrate the endpoints, not the complete evidence ladder.

2 Related Work

Learned dynamics models connect perception to planning in latent state spaces [2, 5, 41, 42]. Object-centric models instead represent entities and interactions explicitly [34, 43, 35]. Benchmarks such as IntPhys, CLEVRER, PLATO, Physion, Physion++, and WorldBench evaluate object persistence, collisions, latent properties, or diagnostic physical concepts [14, 25, 16, 15, 26, 18, 44]. Physics-IQ and related video-model studies further show that perceptual quality and physical validity can diverge [12, 17, 27]. MineOcclude complements these benchmarks with paired occlusion and a continuous causal-FIR-aligned server-state target.

Linear probes can reveal accessible variables in intermediate representations [28–30, 24]. Recent work applies related tools to learned simulators, diffusion models, and video transformers, finding layer- and representation-dependent physical signals [20, 19, 13, 31]. Probe results nevertheless depend on capacity, controls, and model selection [36, 37]. Removal and intervention methods ask the stronger question of whether a representation is used causally [38, 45, 39].

The closest state-probing studies already go beyond output quality. Zhang [20] probe game state in IRIS and DIAMOND and study interventions, while Joseph et al. [31] examine physical representations across depth and spatial structure. Esmati et al. [13] study physical information and interventions in video diffusion models. MineOcclude’s distinction is the matched material/evidence intervention and continuous aligned target, not probing itself or a broader claim about physical-representation mechanisms.

Shin et al. [46] examine hidden-state tracking in a visual Shell Game and distinguish readout-accessible history from state carried across chunks. MineOcclude instead evaluates continuous position in a frozen pretrained checkpoint across five matched material conditions. Its finite recorded windows and condition-specific probes test accessibility, not autonomous long-horizon tracking or causal state maintenance.

3 Methodology

3.1 Paired cohort and evidence ladder

The central graded-visibility study uses 25 new trajectory identities, with 13 left-to-right and 12 right-to-left paths. Each identity is captured in five conditions in counterbalanced order: air (L0), white stained glass (L1), light-blue stained glass (L2), brown stained glass (L3), and stone (L4). Matched captures share the realized object trajectory, camera state, and actions. The five conditions therefore change the scene’s visual evidence without changing the paired object’s path. The apparatus uses one layout, nominal speed 2.0 blocks/s, and position-process noise $q = 0.05$ blocks²/s at $\Delta t = 0.05$ s (20 Hz). The executed update is

$$x_t^{\text{server}} = \text{clip}_{[-8,8]}((x_{t-1}^{\text{server}} + d \cdot 2.0 \Delta t) + \sqrt{q \Delta t} \epsilon_t), \quad \epsilon_t \sim \mathcal{N}(0, 1),$$

where d is motion direction. Identity seeds determine the starting position and innovations; the same realized path is reused across its five conditions. Path hashes check exact pairing. Calibration and seed details are in Appendix A. Figure 1 illustrates the no-wall and opaque endpoints.

There are 125 captures and 22,500 source frames. Each capture contains 180 frames and contributes the same 19 target times, selected by a material-independent counterfactual full-occlusion-extent rule.

78 The rule asks when the complete renderer-derived sheep extent would be hidden by the opaque wall,
 79 rather than selecting rows separately according to each material’s observed decoding performance.
 80 The resulting 2,375 target rows are repeated observations within 25 independent trajectory identities.
 81 No identity is reused from the earlier development study or replaced after outcomes are available.
 82 The supervised target is aligned wall-parallel position, not raw same-tick server state:

$$x_{\text{align}}(t) = \sum_{k=0}^8 h_{d,k} x_{\text{server}}(t - k).$$

83 The frozen direction-specific FIR kernels use only present and past state. They compensate for the
 84 timing difference between server updates and rendered motion, which would otherwise contribute
 85 alignment error to the decoding score. Their coefficients were inherited from earlier calibration,
 86 not fitted to III-A outcomes (Appendix A). Throughout, position error refers to this aligned target.
 87 Renderer geometry independently defines the full object extent and target times.

88 Residual visual evidence (RVE) measures source-image energy attributable to the sheep. For a shared
 89 mask M , let $D = \sum_{u \in M} \|P(u) - A(u)\|_2^2$, where P, A are median present/absent images in linear
 90 light, and let N be the same energy between split-half absent-image medians. The excess energy is
 91 $E = \max(D - N, 0)$. Ratios E_c/E_{L0} are formed within each placement and block, then aggregated
 92 by the median over blocks and nine placements. Let V_c denote this score and $O_c = 1 - V_c$. The
 93 stone measurements yielded zero after noise correction and truncation, not a manually assigned
 94 endpoint. This means no positive residual energy was detected by this procedure, not proof that every
 95 possible cue is absent. Table 1 gives the measured values; Appendix A specifies masks, frames, and
 96 aggregation.

97 RVE is an image-energy coordinate, not a visible-area fraction, an occlusion probability, or MineWorld
 98 confidence. For example, 0.186 does not mean that 18.6% of the object is visible. The glass
 99 conditions differ in color and appearance as well as evidence magnitude. We consequently interpret
 100 the experiment as a controlled material/evidence ladder, not an isolated intervention on transparency
 101 with all other image properties fixed.

Table 1: Central study: measured evidence ladder. Evidence complement is $O_c = 1 - V_c$, not an occlusion probability.

Condition	Material	RVE V_c	Complement O_c	Role
L0	air/no wall	1	0	fully visible control
L1	white stained glass	0.8131	0.1869	high evidence
L2	light-blue stained glass	0.4260	0.5740	intermediate evidence
L3	brown stained glass	0.1857	0.8143	low evidence
L4	stone	0	1	complete occlusion

102 3.2 Frozen MineWorld and feature extraction

103 MineWorld 700M_16f is frozen throughout [11]. Each target uses a teacher-forced forward, meaning
 104 that the model receives recorded images and actions, including the current recorded image, rather
 105 than an autonomous generated continuation. The 16 observations $[t - 15, \dots, t]$ span 0.75 s from
 106 first to last timestamp. Visible decoding can therefore use current sensory evidence directly.

107 The production extraction path calls the adapter separately for each target. Its transformer cache is
 108 refreshed before each call, and token positions start again at zero within the window. No absolute
 109 target time or oracle coordinate is supplied as a model token. Thus, this path cannot access an
 110 observation excluded from the window via a persistent cross-target cache. It is not a test of a
 111 continuously running deployed model. A current-pair feature is the final visual-token slice after
 112 processing the context, not a single-image model input. The same forward supplies layers 4, 10, and
 113 19; storage dimensions and token offsets are in Appendix A.

114 The retained production loader opens source images as RGB, while the adapter applies a red/blue
 115 channel swap and linear resizing to 224×384 . The resulting model input is BGR (blue-green-red),
 116 unlike the checkpoint’s nominal RGB (red-green-blue) convention. Source images for RVE and pixel
 117 ridge remain RGB. This source-code path agrees with the historical audit and frozen configuration;

118 it is not a new runtime channel measurement. Because the intervention includes colored glass, a
 119 channel-order-by-material interaction remains unresolved. Visible accuracy does not rule it out. No
 120 corrected-RGB comparison is reported, so conclusions apply only to the executed BGR configuration;
 121 neither intended-RGB behavior nor channel order as the cause is established.

122 3.3 Readout selection and grouped fitting

123 The earlier paired study selected the representation before the new cohort was evaluated. It screened
 124 layers $\{0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 19\}$ using global mean and coarse 2×6 pooling, then compared
 125 global, coarse, and training-only Best-K refinement at layer 19. Best-K considered $K \in \{1, 4, 16, 64\}$.
 126 The advancing representation was layer-19 coarse pooling. Its selection and qualification results
 127 remain separate from the central study (Appendix B).

128 For the central study, the 14×24 visual grid is averaged into 12 cells, retaining coarse spatial
 129 organization. Flattening these pooled vectors produces 24,576 features. The primary MW_CURRENT
 130 representation uses layer 19. Two prespecified secondary temporal representations use the difference
 131 between current and previous features, also 24,576 dimensions, or previous-first/current concatenation,
 132 49,152 dimensions. Layers 4 and 10 provide secondary depth comparisons; they cannot replace layer
 133 19 after results are inspected.

134 Ridge regression gives a controlled linear readout [47]. Five outer folds hold out complete trajectory
 135 identities, and four grouped inner rotations select $\alpha \in \{0.1, 1, 10, 100, 1000\}$. Standardization,
 136 hyperparameter selection, and fitting use training identities only. Holding related observations
 137 together limits dependence leakage, while inner selection separates tuning from outer evaluation
 138 [48–51].

139 A separate probe is fitted for each condition. The resulting comparison measures how accessible
 140 position is to this readout family within each condition; it does not test whether one fitted decoder
 141 transfers from visible scenes to opaque scenes. All predictions were locked before final statistics,
 142 covering $25 \times 5 \times 19 \times 10 = 23,750$ target/prediction cells for ten models. The layer and representation
 143 were not reselected on this cohort.

144 3.4 Pixel, metadata, and motion comparators

145 The linear current-frame pixel baseline starts from the RGB source image, resizes it with area
 146 interpolation to $48 \times 28 \times 3$, and fits ridge regression to its 4,032 flattened values using the same
 147 grouped tuning procedure. It sees one downsampled image, whereas MineWorld features depend
 148 on a pretrained nonlinear model and up to 16 visual/action pairs. The comparison is therefore not
 149 capacity- or history-matched. It tests this particular current-frame pixel estimator, not all pixel-based
 150 or pixel-history approaches.

151 The training-mean comparator predicts a constant learned from outer-training targets. Direction +
 152 occlusion age fits ordinary least squares to $[1, d, a, da]$, with $d = +1$ for left-to-right motion, $d = -1$
 153 for right-to-left motion, and a the context age. Its coefficients use outer-training rows only. These are
 154 experiment-supplied metadata, not inputs guaranteed to an image-only observer. This tests trajectory
 155 predictability without visual activations, not an equal-information competitor.

156 Last-seen holds the FIR-aligned position from the last counterfactual row that was not fully occluded.
 157 Constant velocity fits position against time over the last ten pre-occlusion positions by ordinary least
 158 squares and extrapolates to the target tick. Both use privileged oracle state history, are identical
 159 across conditions within an identity, and contextualize motion difficulty. They are not evidence that
 160 MineWorld had access to server state. No oracle position is included in the neural or pixel feature
 161 vectors (Table 2).

162 3.5 Metrics and inference

163 For identity i and condition c , let R_{ic} be RMSE across the 19 held-out target rows. Tables report the
 164 mean of these identity-level errors, rather than treating all frames as independent. MAE is aggregated
 165 in the same way. Trajectory R^2 uses the target mean of each held-out path in its denominator; macro
 166 R^2 is the average of these trajectory values.

Predictor	Information supplied
MineWorld current	Recorded visual/action window and pretrained context-dependent features
Pixel ridge	One downsampled source RGB image
Direction + age	Experiment metadata: direction and counterfactual occlusion age
Last-seen	Privileged aligned position at the last non-fully-occluded row
Constant velocity	Ten privileged pre-occlusion aligned positions
Training mean	Outer-training labels only

Table 2: Predictors receive different information. Server-state targets supervise fitting and scoring but never enter the neural or pixel feature vectors.

167 The three primary identity effects are

$$S_i = \text{OLSslope}(R_{ic} \text{ on } O_c), \quad (1)$$

$$D_{\text{partial},i} = \frac{1}{3}(R_{i,L1} + R_{i,L2} + R_{i,L3}) - R_{i,L0}, \quad (2)$$

$$D_{\text{opaque},i} = R_{i,L4} - R_{i,L3}. \quad (3)$$

168 The slope summarizes the complete ladder; the other effects separate the average partial-obstruction
169 change from the final opaque increment. They are fixed contrasts, not comparisons chosen after
170 inspecting the curve.

171 Each one-sided one-sample t-test operates on the 25 identity effects, with statistic $\bar{e}/\text{SE}$, where
172 $\text{SE} = s_e/\sqrt{25}$ and s_e is the sample standard deviation of identity effects, and 24 degrees of
173 freedom, testing a positive mean against zero. Holm correction covers exactly these three primary
174 p-values. Two-sided 95% percentile intervals use 10,000 whole-identity bootstrap samples with seed
175 8258108568742502015 [52]. Each resampled identity carries all conditions and related observations
176 together. The intervals resample locked held-out trajectory summaries with fitted probes fixed; they
177 do not include refitting the pipeline, a new apparatus, or a different checkpoint. The intervals are not
178 Holm-adjusted.

179 Temporal effects measure the reduction in RMSE relative to the current representation, averaged over
180 L1–L4 for each identity. Their two one-sided identity-level t-tests form a separate two-comparison
181 Holm family. Context-age and depth summaries are descriptive. These analyses assume independent
182 identities and a suitable t approximation for the mean effects; overlapping training sets can induce
183 dependence among held-out errors. Identity resampling respects the sampling unit but is not a new
184 independent replication.

185 The central study does not inherit the earlier nine-gate qualification rule or its 49-draw selection-aware
186 permutation test. It reports the prespecified contrasts without an overall experiment pass/fail verdict.
187 Appendix B documents the earlier procedure solely for readout-development provenance.

188 4 Experiments

189 4.1 From readout development to a new cohort

190 The earlier development study reused 25 trajectories from preliminary work. Its matched no-wall and
191 hidden populations each contained 475 rows; a separate 846-row wall-visible control had a different
192 target range and was not a matched effect-size comparison. Layer-19 coarse pooling recovered visible
193 position at 0.109 blocks RMSE without a wall and 0.105 in visible wall-scene frames. Coarse pooling
194 outperformed global averaging at every sampled layer, and Best-K did not improve the advancing
195 representation. The hidden finalist reached 2.071 blocks RMSE and did not meet the development
196 accuracy gates (Appendix B).

197 The central graded-visibility study (Experiment III-A in the records) fixed this readout family, the
198 five conditions, folds, and contrasts before evaluating 25 new identities. Every identity completed
199 every condition. The cohorts are separate at the identity level, not replications across apparatus or
200 checkpoints.

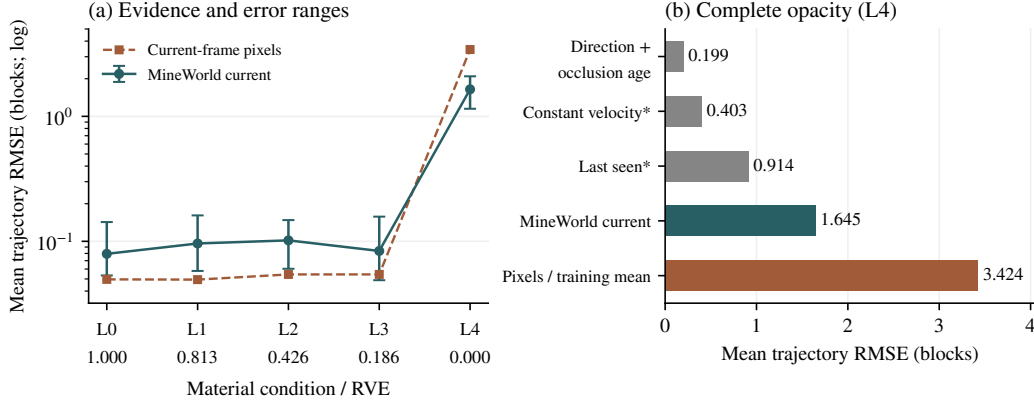


Figure 2: Central graded-visibility study on 25 new identities. (a) Mean trajectory RMSE with reported identity min–max ranges (not confidence intervals); the vertical axis is logarithmic and connecting lines only guide the eye between discrete materials. (b) Opaque-condition comparators on a linear axis. Asterisks mark privileged oracle-history predictors; direction + occlusion age uses experiment metadata. Pixels and training mean share a bar because their L4 errors agree to the displayed precision. Values are reported point estimates, not independent frame-level observations; Table 4 gives intervals for the prespecified paired effects. Pixel and baseline rankings here are descriptive.

4.2 Position decoding across visual evidence

Figure 2 shows the central result. The fixed readout family recovers position accurately in the no-wall condition and through each of the three stained-glass materials. MineWorld RMSEs are 0.079, 0.096, 0.102, and 0.084 blocks for L0–L3, respectively. Their trajectory-macro R^2 values remain between 0.962 and 0.975. The reported stone range (1.150–2.092 blocks) lies wholly above every glass range (all upper bounds below 0.162). This implies higher stone error for every identity, a descriptive deduction from the reported ranges rather than an additional hypothesis test. Under stone, RMSE rises to 1.645 blocks. Table 3 reports the condition-wise values, including MAE and the pixel comparator.

Table 3: Central study: identity-macro condition-wise performance. RMSE and MAE are in blocks. Ranges summarize the 25 reported identity errors, not confidence intervals.

Condition	MW RMSE	MW MAE	MW macro R^2	Pixel RMSE	MW range
L0	0.079	0.065	0.975	0.050	0.053–0.143
L1	0.096	0.080	0.964	0.049	0.058–0.161
L2	0.102	0.082	0.962	0.054	0.060–0.148
L3	0.084	0.069	0.969	0.054	0.049–0.158
L4	1.645	1.187	−9.797	3.424	1.150–2.092

The primary contrasts show a large opaque increment and a much smaller partial-obstruction change (Table 4). Moving from brown glass to stone increases error by 1.561 blocks, with 95% bootstrap interval [1.466, 1.652]. Averaging L1–L3 increases error by 0.0145 blocks relative to no wall, interval [0.0081, 0.0209]. Both effects are positive after the prespecified Holm correction.

The evidence-response slope is also positive, 1.079023 blocks per unit O_c , with interval [1.009511, 1.144293]. However, the slope does not establish a smooth or monotonic decrease in decodability. Brown-glass error is lower than that for the other glass conditions despite its lower measured evidence. Most of the overall slope is associated with the fully opaque endpoint. The opaque increment is about 108 times the average partial increment, a descriptive comparison of effect magnitudes rather than an additional significance test.

Table 4: Central study: three prespecified primary effects. Two-sided 95% identity-bootstrap intervals are not multiplicity-adjusted; Holm correction applies to the three one-sided t-tests. Units are blocks of RMSE for D and blocks of RMSE per unit evidence complement for S .

Estimand	Mean	SE	95% identity-bootstrap CI	Raw p	Holm p
S	1.079	0.0348	[1.010, 1.144]	3.57×10^{-21}	7.14×10^{-21}
D_{partial}	0.0145	0.0033	[0.0081, 0.0209]	9.85×10^{-5}	9.85×10^{-5}
D_{opaque}	1.561	0.0482	[1.466, 1.652]	1.30×10^{-21}	3.89×10^{-21}

4.3 Predictability and pixel comparisons

The linear current-frame pixel baseline has lower error than the MineWorld readout in every L0–L3 condition, at approximately 0.049–0.054 blocks. At L4 its RMSE rises to 3.424 blocks, the reported training-mean level, while the MineWorld readout reaches 1.645 blocks. This is a descriptive advantage over one downsampled current-frame linear comparator; there was no prespecified L4-only significance test of the difference.

MineWorld and pixel ridge differ in history, dimension, and pretrained processing. The opaque-condition ranking therefore cannot distinguish useful history from other feature differences. A matched-history image baseline and controlled history ablations are missing; no superiority over such estimators is established.

The time and motion comparators provide a different test. At L4, direction-and-age achieves RMSE 0.199399 blocks, constant velocity 0.402907, and last-seen 0.913505. All outperform the current MineWorld readout in the reported comparison. These paths are therefore predictable from simple timing and direction information, and from privileged pre-occlusion state history. The baseline inputs differ from the model’s inputs, but the results show that the large opaque error cannot be attributed solely to intrinsically unpredictable target motion.

The direction-and-age result shows that this apparatus supplies strong predictive structure without a neural readout. MineWorld’s advantage over current pixels is therefore not evidence of a better state estimator or a useful memory mechanism. The contribution is the controlled separation of these comparisons, not the neural predictor’s ranking.

In the central cohort, L4 macro $R^2 = -9.797$ despite improvement over the training mean. The constant uses outer-training labels, whereas trajectory R^2 compares squared error with variation around each test path’s own mean. Improving on the former does not imply beating the latter. The test-path mean is a retrospective scoring reference, not an input given to the probe.

4.4 Temporal readouts

Two secondary comparisons test whether simple access to the previous pair improves on the primary representation. Their estimand is the RMSE reduction relative to current features, averaged over L1–L4 within each identity, so positive values indicate improvement. Current-minus-previous is worse, with effect -0.375919 blocks and 95% interval $[-0.394919, -0.358351]$; its one-sided Holm-adjusted p-value is 1. Previous-first two-frame concatenation changes the mean by only 0.000060 blocks, interval $[-0.003808, 0.004229]$, with Holm-adjusted $p = 0.977641$.

The current activation already depends on input history. Differencing can remove stable position information; its p-value tests improvement, not worsening. Concatenation contains the current predictor in its unconstrained linear class, but regularization and finite samples affect the fitted solution. Neither comparison supports average improvement or establishes equivalence. These L1–L4 averages are not L4-only tests and do not cover longer histories, cache-wide readouts, or nonlinear probes.

4.5 Context age and decoder depth

Context age a runs from 1 to 19, or 0.05–0.95 s after the last counterfactual non-fully-occluded row. The last potentially informative pre-occlusion pair is still inside the 16-pair window for $a \leq 15$ and has left it for $a \geq 16$. The presence of a non-fully-occluded frame does not establish localizable

evidence. Window-level localizability and the effects of intervening on history have not been established. Table 5 reports four prespecified descriptive bins. The window schematic in Appendix A explains this boundary; it is not a measured forgetting threshold.

In L4, error is 0.395 and 0.358 blocks in the first two bins, then 2.101 in ages 11–15 and 2.598 in ages 16–19. Errors in the visible and glass conditions remain much lower.

Table 5: Central study: descriptive identity-macro RMSE by context-age bin (blocks).

Age	L0	L1	L2	L3	L4
1–5	0.066	0.090	0.089	0.080	0.395
6–10	0.069	0.097	0.095	0.075	0.358
11–15	0.081	0.085	0.092	0.080	2.101
16–19	0.086	0.091	0.107	0.081	2.598

The boundary analysis uses adjacent windows 12–15 and 16–19, not the broader 11–15 reporting bin. It subtracts the corresponding change averaged over L1–L3 from the L4 change. The interaction is 0.255756 blocks, with 95% whole-identity bootstrap interval [0.060323, 0.448978]. This descriptive analysis lies outside the primary Holm family. Error is already elevated at ages 11–15, while a potentially informative pair remains, but its localizability is unmeasured. At ages 16–19 that pair is excluded by construction. With age, position, and time confounded, neither result diagnoses forgetting in a continuously running model.

The secondary depth panel yields a similar condition pattern at layers 4, 10, and 19 (Table 6). All three support low-error decoding in L0–L3 and much larger errors in L4. These were fixed secondary probes, not additional candidates for choosing the primary result. Their common pattern does not show that the layers use the same mechanism.

4.6 Interpretation and limitations

The protocol separates three questions: how much source-image evidence remains, how accurately a fixed readout family can recover position, and how predictable motion is from non-neural information. In this apparatus, linear decoding remains accurate through glass, while complete opacity produces much larger errors despite strong trajectory predictability. This is a configuration-specific characterization, not a test of downstream planning, control, or causal state maintenance.

New identities separate the central evaluation from readout development but do not establish generalization across layouts, speeds, entities, checkpoints, or renderers. The independent sample is 25 identities. A second physical condition and intended-RGB replication are needed before extending the empirical conclusion.

The shared legacy BGR path supports within-apparatus comparisons but cannot exclude a preprocessing-by-occlusion interaction or establish intended-RGB behavior. Glass color and appearance also change with evidence magnitude, and separately fitted probes do not test decoder transfer. Longer histories, cache-wide and nonlinear readouts remain untested. Neither the earlier selection-aware null nor the new identity-level tests establish causal memory, causal forgetting, calibrated beliefs, or absence of object state from MineWorld as a whole.

5 Conclusion

MineOcclude provides a paired protocol for comparing visual evidence, linear position decoding, and motion predictability. On 25 new identities, a readout family fixed from development achieves low error through glass and substantially higher error behind stone. Metadata and oracle-history predictors remain more accurate under opacity. The study establishes this pattern for reset-window MineWorld features under the executed BGR path, not a general limit on hidden-state representation. Corrected-input and matched-history experiments, followed by a second physical condition, are required to extend that conclusion.

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477 A Central-study measurement and implementation details

478 These details describe the committed III-A implementation and frozen authorities, not new inference
 479 or a rerun of statistics. The implementation inspection used the production extraction entypoint, the
 480 shared adapter executor, and the cohort, FIR, RVE, and fold modules. Source-level consistency does
 481 not independently verify every tensor in the completed run.

482 A.1 Motion, target alignment, and input window

483 The lane spans $x \in [-8, 8]$ at $(y, z) = (4, 6)$, with a locked camera at eye position
 484 $(0.5, 5.6200000048, -5.5)$, zero yaw/pitch/roll, and 70° field of view. The wall occupies $[-3, 4] \times$
 485 $[4, 7] \times [2, 3]$ in (x, y, z) . Minecraft 1.16.5 renders 640×360 frames. Conditions preserve the camera,
 486 realized motion, lighting, and action schedule; they differ in wall material.

487 The generator uses `NumPy default_rng(seed).standard_normal(180)`, scaled by $\sqrt{q\Delta t}$. It
 488 accumulates drift before noise in binary64 and clamps each step, not a vectorized cumulative sum. The
 489 identity seed also seeds innovations. Starting position is drawn from its direction-specific range using
 490 `default_rng(seed XOR 0xE3E3E3E3)`. The deterministic candidate stream is qualified before
 491 outcomes; the numerical identity roster and start states belong to the cohort manifest. Recorded action
 492 rows are converted to 11 tokens per pair. This revision does not establish that every action component
 493 was constant or contains no timing cue; exact action-stream verification requires the capture records.

494 The following raw FIR taps, ordered by lag $0, \dots, 8$, are the frozen recovered values:

	Lag	Left-to-right	Right-to-left
	0	0.0552210582	0.0155128136
	1	0.0955547802	0.1420986717
	2	0.1729033607	0.3819949723
495	3	0.2562377030	0.2529919934
	4	0.1725945773	0.1314006058
	5	0.1502955787	0.0092585384
	6	0.0283805770	0.0336603539
	7	0.0327627758	0.0330820509
	8	0.0360495890	0.0000000000

496 At load time, each vector is divided by its `math.fsum`; filtered sums also use `math.fsum`. The recovery
 497 receipt reconstructs the taps from the earlier materialized stage-2 aligned labels and regenerated
 498 paths, checking 50 path hashes and 950 labels. This recovers inherited measurement calibration, not a
 499 fit to III-A decoding outcomes. The raw left-to-right vector sums to 0.9999999999 owing to decimal
 500 rounding, so normalization is part of the implementation.

	Schematic target age	Recorded window relative to last non-full row t_0	t_0 included?
501	$a = 1$ (0.05 s)	$[t_0 - 14, \dots, t_0 + 1]$	Yes
	$a = 15$ (0.75 s)	$[t_0, \dots, t_0 + 15]$	Yes
	$a = 16$ (0.80 s)	$[t_0 + 1, \dots, t_0 + 16]$	No

502 The timeline is schematic, not a plot of measured forgetting. The production adapter calls
 503 `refresh_kvcache()` before the forward; positions are consecutive context-relative indices. The
 504 source loader converts PNGs to RGB and the shared preprocessor swaps red and blue. The pilot
 505 entypoint uses a different loader, so the production path, rather than the pilot filename or comments
 506 alone, determines this description.

507 A.2 Evidence calibration and grouped readouts

508 Calibration uses two present/absent blocks (P-A-A-P and A-P-P-A), with at least eight settled
 509 frames per state per block, at nine lane positions $x = -2, -1.5, \dots, 2$, $y = 4$, $z = 6$. These are
 510 calibration placements, not the 25 experimental identities. Source values $s = v/255$ are linearized
 511 as $s/12.92$ for $s \leq 0.04045$ and $((s + 0.055)/1.055)^{2.4}$ otherwise. Coordinate-wise medians form
 512 each state's robust image.

For each placement, the mask threshold is the maximum squared channel-summed residual over pixels and blocks between two L0 absent-half medians. Pixels with L0 present/absent residual strictly above this threshold are unioned across blocks and dilated by a two-pixel square neighborhood. This one mask is shared across materials. Absent frames are split by ascending capture index into equal first/last halves; an odd middle frame is omitted. Ratios of nonnegative excess energies are formed within block and placement before taking the median over the two blocks and then over the nine placements. The calibration minimum is a protocol requirement; the exact realized image inventory is needed to inspect individual captures.

Each pair contains 336 visual and 11 action tokens. The 16-pair sequence has 5,552 tokens and hidden width 2,048. Previous and current visual offsets are [4858:5194) and [5205:5541); action tokens are excluded from readout vectors. The reported 750 float16 arrays each have shape [19, 336, 2048] and file size 26,148,992 bytes, approximately 18.26 GiB together. All 2,375 planned production forwards are reported complete.

Outer folds sort identities by (seed, identity ID) within direction and allocate five identities per fold with left/right counts (3, 2), (2, 3), (3, 2), (2, 3), (3, 2). Four deterministic inner rotations are constructed inside each outer training set. Exact membership is bound by the cohort/fold manifests; every condition and row of an identity stays together. All training-dependent transforms and ridge selection remain inside these splits.

A.3 Additional central results and access

Table 6: Central study: current-readout identity-macro RMSE across the frozen depth panel (blocks).

Layer	L0	L1	L2	L3	L4
4	0.076	0.108	0.117	0.094	1.698
10	0.079	0.100	0.105	0.083	1.648
19	0.079	0.096	0.102	0.084	1.645

Capture and extraction elapsed times reported for III-A are 2,975.5 s (49.6 min) and 4,730.8 s (78.8 min), respectively. They are stage timings, not total project compute. The available records do not establish complete III-A hardware, fitting, and development costs.

The compact unit of statistical reproduction is the held-out prediction package, not the activation corpus. The reported prediction lock is 4,833,894 bytes. It must be released with its original seal, identity/condition/row/model keys, predictions and aligned targets, outer/inner folds, RVE coordinates, row-to-age mapping, analysis configuration and seeds, executed analysis source revision, environment, and expected tables and intervals. Reconstruction must start from individual predictions, verify complete paired cells and hashes, and regenerate identity-level errors, contrasts, Holm adjustments, and bootstrap intervals without model inference or probe refitting.

The actual lock and matched central captures were unavailable for this revision. The figures therefore use reported summaries; no independent reconstruction of their distributions or intervals is claimed. The inspected preparation entripoint accepts separate per-identity estimands, and its context-stage branch emits metadata rather than the reported numerical interaction. That entripoint alone is not a complete reproduction of the paper. The exact executed analysis and estimand-construction code must accompany the lock. The anonymous repository is a project entripoint; a complete statistics snapshot has not been verified. The reproducibility checklist remains negative until these files are supplied and successfully checked.

B Readout-development provenance

The earlier paired study (Experiment II in the execution records) selected the representation subsequently fixed for the central graded-visibility study (Experiment III-A). It reused 25 trajectories and tested 11 layers with global and coarse pooling, followed by layer-19 refinement. Its nine-gate decision below is not the decision rule for the new-cohort study.

For identity i , $\Delta_{A,i}$ is training-mean RMSE minus probe RMSE. The five fold means were 0.38572, 0.42825, 0.44047, 0.38606, and 0.35533. Direction means were 0.37864 and 0.42141; the avail-

Gate	Earlier-study requirement	Finalist
G1	Provenance, cohort, target, implementation, and leakage checks	Pass
G2	Mean $\Delta_A > 0$ and bootstrap lower bound > 0	Pass
G3	Macro $R^2 > 0$ and bootstrap lower bound > 0	Not met
G4	RMSE ≤ 0.50 blocks	Not met
G5	$\geq 4/5$ fold means > 0 ; every fold mean > -0.10	Pass
G6	Both direction means > 0	Pass
G7	$\geq 4/5$ phase means ≥ 0 ; every phase mean > -0.10	Pass
G8	Both visible controls qualify	Pass
G9	Selection-aware $p_A \leq 0.05$	Pass

Table 7: Earlier-study qualification. Stability thresholds apply to group means of Δ_A , in blocks. The central study uses the three effect tests in subsection 3.5, not this ledger.

able record does not identify their displayed left/right order. Phase means for phases 0, 3, 6, 9, 12 were 1.04799, 0.64187, 0.30272, -0.00659 , and 0.00985 . These values establish G5–G7 without introducing new thresholds.

The final three-candidate refinement selected layer-19 coarse pooling with mean $\Delta_A = 0.3991669$. Each of 49 null draws permuted hidden activation blocks within direction-by-phase groups and reran the reported selection procedure, leaving targets, folds, baselines, age, and visible controls fixed [53]. The null mean was 0.3845741 and its maximum 0.3989598, only 0.0002071 blocks below the observation. The add-one p-value was $(1 + 0)/(49 + 1) = 0.02$, its minimum resolution [54]. The finalist’s macro $R^2 = -13.8263$ and RMSE 2.0711 nevertheless failed G3 and G4. The summary identifies three layer-19 refinement candidates but does not establish that the earlier 22-cell layer screen was rerun for each null. We therefore do not claim adjustment for that entire earlier search. These statistics concern development only, not the central three-effect tests.

Extraction for that study involved 1,796 target rows and 11 tensors per forward, giving 19,756 arrays of shape [336, 2048] in little-endian float16. The tensor payload totaled 27,189,313,536 bytes (25.32 GiB). The project describes extraction code, configurations, inventories, and integrity manifests separately from this raw corpus. Complete raw-data access and an executable public snapshot still require verification; hashes alone do not establish reproducibility.

The V1 audit had ten applicable checks that matched and four not-applicable entries. It checked source/model identities, paths, targets, tensor integrity, selected features, folds, and the screen/refinement, including decisive metrics. It did not independently recompute the final selection null, bootstrap, visible controls, or all gate arithmetic. One not-applicable explanation incorrectly stated that the final artifact did not exist; the artifact does exist. The audit therefore does not certify every final statistical calculation.

The historical preprocessing identifier was `person-a-bgr2rgb-inter-linear-224x384-v1.0`, with 224×384 linear resizing. The audited model input was BGR despite that identifier. A metadata repair aligned the recorded identity with the executed path and did not change scientific results; the preceding metadata version had produced no tensors. The central study’s authority retained this path. This provenance supports the configuration-specific scope stated in subsection 3.2 and subsection 4.6.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and introduction center the graded-visibility study on 25 new identities. They distinguish within-condition linear decodability from causal memory or absent model state, and restrict conclusions to the tested checkpoint and legacy preprocessing. Sections 4.2–4.6 report the central results and limitations.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 4.6 and the methodology discuss the small single-apparatus cohort, retained BGR preprocessing, material appearance changes, separately fitted condition-specific probes, limited readout families, and statistical assumptions. The earlier development cohort is distinguished from the new identities.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

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Justification: The paper presents an empirical study and does not introduce theoretical results, theorems, or formal proofs.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [No]

Justification: The paper discloses the principal design, kernels, measurement rule, grouped fits, and tests. However, the exact realized action streams and a complete runnable central-study prediction/configuration snapshot were not available for verification. The supplement distinguishes these missing artifacts from the disclosed methods.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: An anonymous project link is provided, but a complete downloadable central prediction/configuration snapshot has not been verified. Complete capture and activation corpora are not included in the reviewed materials. The earlier activation corpus alone is 25.32 GiB; the central study additionally produced 750 feature arrays. Full data access or regeneration is required.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Sections 3.1–3.5 describe cohort construction, the fixed primary representation, grouped splits, ridge alpha selection, preprocessing, baseline inputs, and inferential units. Section 4 separates the central study from the earlier development protocol.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Section 3.5 specifies the identity-level one-sided t-tests, three-test primary and two-test temporal Holm families, and two-sided 95 percent percentile intervals from 10,000 whole-identity resamples. The primary-effects table and Sections 4.2–4.5 distinguish inferential effects from descriptive summaries. Figure 2 shows mean trajectory errors and reported identity min–max ranges, explicitly distinguished from confidence intervals. Bootstrap intervals keep the fitted probes fixed.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [No]

Justification: The appendix reports the earlier extraction workload and 25.32 GiB tensor payload. The central study reports 2,375 forwards, 750 arrays, shapes, and file sizes, and the supplement reports capture/extraction stage timings, but complete hardware, fitting, and development costs are not provided.

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Justification: MineWorld 700M_16f is credited and used under the MIT License, and Fabric API 0.42.0+1.16 is used under Apache-2.0. Minecraft Java Edition 1.16.5 is used without redistributing game files and in accordance with the Mojang/Microsoft EULA and Usage Guidelines.

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706 Question: Does the paper describe the usage of LLMs if it is an important, original, or
707 non-standard component of the core methods in this research? Note that if the LLM is used
708 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
709 scientific rigor, or originality of the research, declaration is not required.

710 Answer: [N/A]

711 Justification: No LLM was an experimental component or used to generate data, labels,
712 or scientific results. LLM assistance was limited to manuscript preparation and coding or
713 debugging support, while the authors reviewed and validated the implementation and all
714 scientific decisions.